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Course: SCSM2223 CROSS-PLATFORM APPLICATION DEVELOPMENT

Section: 04

(A) Screenshots of working UI

Employee form and list: List shows the active and inactive employees

The screenshot displays the 'Employee Directory' interface. On the left, the 'Add New Employee' form is visible, featuring fields for Employee ID, Full Name, Email, Department, Position, Hire Date, and Salary (RM). Below the form is a radio button to toggle between 'Active employee' and 'Inactive employee'. On the right, a table lists existing employees with columns for EMP ID, NAME, DEPARTMENT, POSITION, HIRE DATE, SALARY, STATUS, and ACTIONS. The table includes 8 employees, with their status indicated by a green 'Active' or red 'Inactive' label.

EMP ID	NAME	DEPARTMENT	POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
EMP001	Ahmad Fais bin Zulkarna	IT	Software Engineer	15 May 2021	RM 5,500.00	Active	Edit Delete
EMP002	Haz Hilalqah binl Ruzlan	HR	HR Executive	1 Jul 2020	RM 4,200.00	Active	Edit Delete
EMP003	Lim Wei Jian	Finance	Financial Analyst	20 Nov 2019	RM 6,000.00	Active	Edit Delete
EMP004	Siti Aminah binti Yusuf	IT	System Administrator	10 Jan 2022	RM 4,800.00	Active	Edit Delete
EMP005	Rajesh u/l Subramaniam	Finance	Accounts Executive	12 May 2020	RM 4,500.00	Inactive	Edit Delete
EMP006	Harul Ain binti Hanuman	HR	Recruitment Specialist	14 Feb 2023	RM 3,800.00	Active	Edit Delete
EMP007	Cheng Kah Meng	IT	DevOps Engineer	20 May 2020	RM 7,200.00	Active	Edit Delete
EMP008	Teng Yu Shan	IT	Data Analyst	11 May 2025	RM 6,000.00	Active	Edit Delete

Click Add Employee with empty form:

This screenshot shows the 'Add New Employee' form with all fields empty. The form includes input fields for Employee ID, Full Name, Email, Department, Position, Hire Date, and Salary (RM). A radio button is set to 'Active employee'. The 'Add Employee' button is at the bottom of the form.

Add new employee: Fill the form and click 'Add Employee'

This screenshot shows the 'Add New Employee' form filled out with the following details: Employee ID: EMP009, Full Name: Rajesh u/l Subramaniam, Email: rajesh.sub@company.com, Department: IT, Position: Data Analyst, Hire Date: 11/05/2025, Salary (RM): 6000.00. The 'Add Employee' button is highlighted. To the right, the employee list is updated to include the new employee, EMP009, with a status of 'Active'.

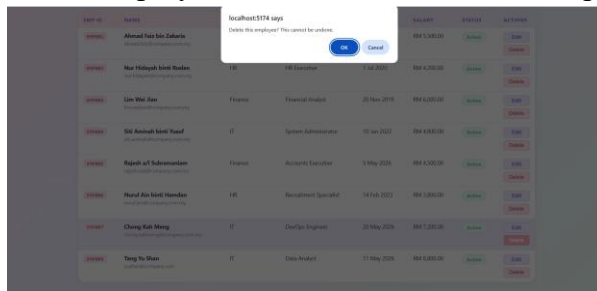
EMP ID	NAME	DEPARTMENT	POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
EMP001	Ahmad Fais bin Zulkarna	IT	Software Engineer	15 May 2021	RM 5,500.00	Active	Edit Delete
EMP002	Haz Hilalqah binl Ruzlan	HR	HR Executive	1 Jul 2020	RM 4,200.00	Active	Edit Delete
EMP003	Lim Wei Jian	Finance	Financial Analyst	20 Nov 2019	RM 6,000.00	Active	Edit Delete
EMP004	Siti Aminah binti Yusuf	IT	System Administrator	10 Jan 2022	RM 4,800.00	Active	Edit Delete
EMP005	Rajesh u/l Subramaniam	Finance	Accounts Executive	12 May 2020	RM 4,500.00	Inactive	Edit Delete
EMP006	Harul Ain binti Hanuman	HR	Recruitment Specialist	14 Feb 2023	RM 3,800.00	Active	Edit Delete
EMP007	Cheng Kah Meng	IT	DevOps Engineer	20 May 2020	RM 7,200.00	Active	Edit Delete
EMP008	Teng Yu Shan	IT	Data Analyst	11 May 2025	RM 6,000.00	Active	Edit Delete
EMP009	Rajesh u/l Subramaniam	IT	Data Analyst	11 May 2025	RM 6,000.00	Active	Edit Delete

Edit employee details: Click 'Edit' form the employee list, edit and click 'Update'

This screenshot shows the 'Edit Employee' form, which is accessed by clicking the 'Edit' button for an employee in the list. The form is pre-filled with the details of the selected employee: Employee ID: EMP005, Full Name: Rajesh u/l Subramaniam, Email: rajesh.sub@company.com, Department: Finance, Position: Accounts Executive, Hire Date: 12/05/2020, Salary (RM): 4500.00. The 'Update' button is highlighted. To the right, the employee list remains the same as in the previous screenshot.

EMP ID	NAME	DEPARTMENT	POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
EMP001	Ahmad Fais bin Zulkarna	IT	Software Engineer	15 May 2021	RM 5,500.00	Active	Edit Delete
EMP002	Haz Hilalqah binl Ruzlan	HR	HR Executive	1 Jul 2020	RM 4,200.00	Active	Edit Delete
EMP003	Lim Wei Jian	Finance	Financial Analyst	20 Nov 2019	RM 6,000.00	Active	Edit Delete
EMP004	Siti Aminah binti Yusuf	IT	System Administrator	10 Jan 2022	RM 4,800.00	Active	Edit Delete
EMP005	Rajesh u/l Subramaniam	Finance	Accounts Executive	12 May 2020	RM 4,500.00	Active	Edit Delete
EMP006	Harul Ain binti Hanuman	HR	Recruitment Specialist	14 Feb 2023	RM 3,800.00	Active	Edit Delete
EMP007	Cheng Kah Meng	IT	DevOps Engineer	20 May 2020	RM 7,200.00	Active	Edit Delete
EMP008	Teng Yu Shan	IT	Data Analyst	11 May 2025	RM 6,000.00	Active	Edit Delete

Delete employee: Click ‘Delete’ from the employee list, select ‘OK’



The screenshot shows a table of employees with the following columns: EMP ID, NAME, DEPARTMENT, POSITION, HIRE DATE, SALARY, STATUS, and ACTIONS. The table contains 10 rows of employee data.

EMP ID	NAME	DEPARTMENT	POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
EMP001	Ahmad Fala bin Zakaria	IT	Software Engineer	15 Mar 2021	RM 5,500.00	Active	Edit Delete
EMP002	Nur Hidayah binti Ruslan	HR	HR Executive	1 Jul 2020	RM 4,200.00	Active	Edit Delete
EMP003	Lim Wei Jian	Finance	Financial Analyst	20 Nov 2019	RM 6,000.00	Active	Edit Delete
EMP004	Siti Aminah binti Yusoff	IT	System Administrator	10 Jan 2022	RM 4,800.00	Active	Edit Delete
EMP005	Rajesh a/l Subramaniam	Finance	Accounts Executive	5 May 2026	RM 4,500.00	Active	Edit Delete
EMP006	Nurul Ain binti Hameedan	HR	Recruitment Specialist	14 Feb 2023	RM 3,800.00	Active	Edit Delete
EMP007	Cheng Kah Meng	IT	CloudOps Engineer	20 May 2026	RM 7,200.00	Active	Edit Delete
EMP008	Teng Yu Shan	IT	Data Analyst	11 May 2026	RM 8,000.00	Active	Edit Delete

Search employee: Type name, id, email, department or position to find employee

The screenshot shows the "Add New Employee" form. It includes fields for Employee ID, Full Name, Email, Department, Position, Hire Date, and Salary (RM). There is a checkbox for "Active employee" and an "Add Employee" button. Below the form is a search bar with the text "Teng" and a dropdown menu showing "Teng Yu Shan".

The screenshot shows the "Add New Employee" form with a search bar containing the text "Siti". The dropdown menu shows "Siti Aminah binti Yusoff".

Sort employee: Can sort by name, hire date, salary or department in ascending/descending

The screenshot shows a table of employees sorted by name in ascending order. The columns are EMP ID, NAME, DEPARTMENT, POSITION, HIRE DATE, SALARY, STATUS, and ACTIONS.

EMP ID	NAME	DEPARTMENT	POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
EMP001	Ahmad Fala bin Zakaria	IT	Software Engineer	15 Mar 2021	RM 5,500.00	Active	Edit Delete
EMP003	Lim Wei Jian	Finance	Financial Analyst	20 Nov 2019	RM 6,000.00	Active	Edit Delete
EMP002	Nur Hidayah binti Ruslan	HR	HR Executive	1 Jul 2020	RM 4,200.00	Active	Edit Delete
EMP006	Nurul Ain binti Hameedan	HR	Recruitment Specialist	14 Feb 2023	RM 3,800.00	Active	Edit Delete
EMP005	Rajesh a/l Subramaniam	Finance	Accounts Executive	5 May 2026	RM 4,500.00	Active	Edit Delete
EMP004	Siti Aminah binti Yusoff	IT	System Administrator	10 Jan 2022	RM 4,800.00	Active	Edit Delete
EMP008	Teng Yu Shan	IT	Data Analyst	11 May 2026	RM 8,000.00	Active	Edit Delete

The screenshot shows a table of employees sorted by hire date in descending order. The columns are EMP ID, NAME, DEPARTMENT, POSITION, HIRE DATE, SALARY, STATUS, and ACTIONS.

EMP ID	NAME	DEPARTMENT	POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
EMP008	Teng Yu Shan	IT	Data Analyst	11 May 2026	RM 8,000.00	Active	Edit Delete
EMP005	Rajesh a/l Subramaniam	Finance	Accounts Executive	5 May 2026	RM 4,500.00	Active	Edit Delete
EMP006	Nurul Ain binti Hameedan	HR	Recruitment Specialist	14 Feb 2023	RM 3,800.00	Active	Edit Delete
EMP004	Siti Aminah binti Yusoff	IT	System Administrator	10 Jan 2022	RM 4,800.00	Active	Edit Delete
EMP001	Ahmad Fala bin Zakaria	IT	Software Engineer	15 Mar 2021	RM 5,500.00	Active	Edit Delete
EMP002	Nur Hidayah binti Ruslan	HR	HR Executive	1 Jul 2020	RM 4,200.00	Active	Edit Delete
EMP003	Lim Wei Jian	Finance	Financial Analyst	20 Nov 2019	RM 6,000.00	Active	Edit Delete

The screenshot shows a table of employees sorted by salary in descending order. The columns are EMP ID, NAME, DEPARTMENT, POSITION, HIRE DATE, SALARY, STATUS, and ACTIONS.

EMP ID	NAME	DEPARTMENT	POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
EMP008	Teng Yu Shan	IT	Data Analyst	11 May 2026	RM 8,000.00	Active	Edit Delete
EMP003	Lim Wei Jian	Finance	Financial Analyst	20 Nov 2019	RM 6,000.00	Active	Edit Delete
EMP001	Ahmad Fala bin Zakaria	IT	Software Engineer	15 Mar 2021	RM 5,500.00	Active	Edit Delete
EMP004	Siti Aminah binti Yusoff	IT	System Administrator	10 Jan 2022	RM 4,800.00	Active	Edit Delete
EMP005	Rajesh a/l Subramaniam	Finance	Accounts Executive	5 May 2026	RM 4,500.00	Active	Edit Delete
EMP002	Nur Hidayah binti Ruslan	HR	HR Executive	1 Jul 2020	RM 4,200.00	Active	Edit Delete
EMP006	Nurul Ain binti Hameedan	HR	Recruitment Specialist	14 Feb 2023	RM 3,800.00	Active	Edit Delete

The screenshot shows a table of employees sorted by department in ascending order. The columns are EMP ID, NAME, DEPARTMENT, POSITION, HIRE DATE, SALARY, STATUS, and ACTIONS.

EMP ID	NAME	DEPARTMENT	POSITION	HIRE DATE	SALARY	STATUS	ACTIONS
EMP003	Lim Wei Jian	Finance	Financial Analyst	20 Nov 2019	RM 6,000.00	Active	Edit Delete
EMP005	Rajesh a/l Subramaniam	Finance	Accounts Executive	5 May 2026	RM 4,500.00	Active	Edit Delete
EMP002	Nur Hidayah binti Ruslan	HR	HR Executive	1 Jul 2020	RM 4,200.00	Active	Edit Delete
EMP006	Nurul Ain binti Hameedan	HR	Recruitment Specialist	14 Feb 2023	RM 3,800.00	Active	Edit Delete
EMP001	Ahmad Fala bin Zakaria	IT	Software Engineer	15 Mar 2021	RM 5,500.00	Active	Edit Delete
EMP004	Siti Aminah binti Yusoff	IT	System Administrator	10 Jan 2022	RM 4,800.00	Active	Edit Delete
EMP008	Teng Yu Shan	IT	Data Analyst	11 May 2026	RM 8,000.00	Active	Edit Delete

(B) How my code satisfies each of the four Chapter 8 learning outcomes

LO1. Connect a Vue 3 application to an external REST API and manage the request / response lifecycle.

The Vue 3 frontend connects to the Express REST API through a centralised Axios service module located at *employeeApi.js* using a baseUrl: 'http://localhost:3001'. Vue application never communicates with the MySQL database, which data flows through the REST API.

LO2. Configure and use the Axios HTTP client, including interceptors for logging and centralised error handling.

The Axios HTTP client is configured in *src/api/employeeApi.js* with baseUrl: 'http://localhost:3001'. Request interceptor logs every outgoing request to the browser console while response interceptor detects the error and convert into message. In *App.vue*, *async*, *await*, *try* and *catch* are used with the HTTP calls.

LO3. Build Vue forms that capture user input using v-model, validate input with custom rules, and communicate results back to a parent component via events.

In *EmployeeForm.vue*, *v-model* is used with *.trim* to strip whitespace and *.number* modifier to ensure numeric type conversion. *v-if* also used to display message when there is invalid input.

LO4. Build a small Express REST API that performs full CRUD against a MySQL database using prepared statements, including server-side search and sort via query parameters.

In *index.js*, every SQL query uses '?' placeholders instead of putting the value directly: *'SELECT * FROM employees WHERE id = ?'*. GET */employees* supports server-side LIKE search and ORDER BY sorting, such as vue sends *?q=keyword* to the API and the search happens in the MySQL.

(C) Challenges faced and how to resolve them

1. Two servers have to run at the same time

Two different terminal have to be used to run the command: *npm run server*, *npm run dev*

2. Active employee checkbox mismatch the field type

Adding *console.log('active received:', active)*

3. Action buttons appeared at different vertical positions

Setting *vertical-align: middle* and use *flex direction: column* to ensure the buttons are stacked.